

Smoker Drinker or Both

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Features

- Sex
- Age
- Height
- Weight
- Eyesight (left)
- Eyesight (right)
- Hearing (left)
- Hearing (right)
- Systolic blood pressure
- Diastolic blood pressure
- BLDS or FSG (fasting blood glucose)
- SGOT (Glutamate-oxaloacetate transaminase)
- AST (Aspartate transaminase)
- Total cholesterol
- HDL cholesterol
- LDL cholesterol
- Triglyceride
- Hemoglobin
- Protein in urine
- Serum (blood) creatinine
- ALT (Alanine transaminase)
- Y-glutamyl transpeptidase

Response variables

- Smoking State
- Drinking State

Classification Problem

Before Preprocessing

sex	age	height	weight	waistline	sight_left	sight_right	hear_left	hear_right	SBP	DBP	BLDS	tot_chole	HDL_chole	LDL_chole	triglyceri	hemoglobi	urine_prot	serum_cre	SGOT_AST	SGOT_ALT	gamma_G	SMK_stat_	DRK_YN
Male	35	170	75	90	1	1	1	1	120	80	99	193	48	126	92	17.1	1	1	21	35	40	1	Y
Male	30	180	80	89	0.9	1.2	1	1	130	82	106	228	55	148	121	15.8	1	0.9	20	36	27	3	N
Male	40	165	75	91	1.2	1.5	1	1	120	70	98	136	41	74	104	15.8	1	0.9	47	32	68	1	N
Male	50	175	80	91	1.5	1.2	1	1	145	87	95	201	76	104	106	17.6	1	1.1	29	34	18	1	N
Male	50	165	60	80	1	1.2	1	1	138	82	101	199	61	117	104	13.8	1	0.8	19	12	25	1	N
Male	50	165	55	75	1.2	1.5	1	1	142	92	99	218	77	95	232	13.8	3	0.8	29	40	37	3	Y
Female	45	150	55	69	0.5	0.4	1	1	101	58	89	196	66	115	75	12.3	1	0.8	19	12	12	1	N
Male	35	175	65	84.2	1.2	1	1	1	132	80	94	185	58	107	101	14.4	1	0.8	18	18	35	3	Y
Male	55	170	75	84	1.2	0.9	1	1	145	85	104	217	56	141	100	15.1	1	0.8	32	23	26	1	Y
Male	40	175	75	82	1.5	1.5	1	1	132	105	100	195	60	118	83	13.9	1	0.9	21	38	16	2	Y
Male	45	155	55	79.2	1	1	1	1	118	70	90	183	42	130	55	12.9	1	0.8	19	14	19	1	N
Male	65	155	75	98	1.2	9.9	1	1	109	69	137	115	31	57	137	16.5	1	1.3	48	51	42	2	N
Female	55	150	55	72.3	1.2	0.9	1	1	130	80	106	183	51	89	214	13.1	1	0.8	20	20	31	1	N
Male	30	175	75	88	1.2	1.2	1	1	118	72	82	200	55	129	77	15.7	1	0.9	13	16	19	1	Y
Female	30	160	50	76	0.9	1	1	1	129	77	79	205	53	108	219	14.5	1	0.6	21	24	58	1	N
Male	40	170	65	80	1	1	1	1	113	72	104	113	44	62	35	16	2	0.9	15	10	12	1	N
Female	25	160	65	73	1.2	0.9	1	1	126	78	96	148	54	82	60	12.3	1	0.5	12	7	9	1	Y
Male	25	170	65	78	1.2	1.2	1	1	119	67	100	147	51	85	54	14.8	1	0.9	33	17	13	1	Y
Male	50	170	85	99	0.7	0.8	1	1	121	74	99	180	43	103	169	14.4	1	1.2	41	51	60	1	Y
Male	60	165	60	85	0.3	0.7	1	1	120	85	105	197	42	111	222	15.2	1	1	19	20	48	3	Y
Female	35	170	50	67	1	0.8	1	1	111	65	88	174	66	98	46	12.1	1	0.9	19	13	18	1	Y
Male	25	175	65	82	1.5	1.5	1	1	130	76	95	207	62	134	57	16.7	1	1.2	28	30	32	3	Y
Female	45	155	50	62	0.5	0.7	1	1	109	64	111	170	39	64	333	12.5	1	0.7	17	21	60	1	N
Male	40	165	75	92	1	1.5	1	1	110	70	102	195	57	113	125	16.1	1	1	29	31	33	2	Y
Female	20	160	55	79	1.2	1.5	1	1	110	70	87	156	59	86	55	12.1	1	0.8	16	11	15	1	Y
Male	60	170	75	87	0.7	0.7	1	1	119	80	83	154	35	104	150	15.9	1	0.7	31	29	76	3	Y

After Preprocessing

age	height	weight	waistline	sight_left	sight_right	SBP	DBP	BLDS	tot_chole	HDL_chole	LDL_chole	triglyceride	hemoglobi	serum_cre	SGOT_AST	SGOT_ALT	gamma_G	sex	urine_prot	hear_left	hear_right	DRK_YN	SMK_stat_
-0.88951	0.835874	0.93621	0.739781	0.031629	0.035668	-0.16726	0.399153	-0.05891	-0.06614	-0.51842	0.361643	-0.39279	1.810919	0.290374	-0.21237	0.351404	0.056791	0	1	1	1	1	0
-1.24209	1.913117	1.335755	0.655395	-0.1334	0.36637	0.520349	0.601391	0.230586	0.839184	-0.11235	0.975435	-0.10902	0.990693	0.08227	-0.25494	0.389415	-0.20102	0	1	1	1	0	1
-0.53694	0.297252	0.93621	0.824167	0.36169	0.862423	-0.16726	-0.61203	-0.10027	-1.54053	-0.92449	-1.08914	-0.27537	0.990693	0.08227	0.894324	0.237373	0.612081	0	1	1	1	0	0
0.168215	1.374495	1.335755	0.824167	0.856782	0.36637	1.551763	1.106985	-0.22434	0.14079	1.105852	-0.25215	-0.2558	2.126391	0.498477	0.128151	0.313394	-0.37951	0	1	1	1	0	0
0.168215	0.297252	-0.26243	-0.10408	0.031629	0.36637	1.070436	0.601391	0.023803	0.089058	0.235705	0.110547	-0.27537	-0.27119	-0.12583	-0.2975	-0.52284	-0.24069	0	1	1	1	0	0
0.168215	0.297252	-0.66197	-0.52601	0.36169	0.862423	1.34548	1.612579	-0.05891	0.58052	1.163862	-0.50324	0.977116	-0.27119	-0.12583	0.128151	0.541456	-0.0027	0	3	1	1	1	1
-0.18436	-1.31861	-0.66197	-1.03232	-0.79352	-0.95644	-1.47372	-1.82546	-0.47248	0.011458	0.525754	0.054748	-0.55913	-1.21761	-0.12583	-0.2975	-0.52284	-0.4985	1	1	1	1	0	0
-0.88951	1.374495	0.13712	0.250343	0.36169	0.035668	0.65787	0.399153	-0.26569	-0.27307	0.061676	-0.16845	-0.30472	0.107372	-0.12583	-0.34007	-0.29477	-0.04237	0	1	1	1	1	1
0.520791	0.835874	0.93621	0.233466	0.36169	-0.12968	1.551763	0.904747	0.147873	0.554654	-0.05434	0.780138	-0.31451	0.549032	-0.12583	0.255846	-0.10472	-0.22085	0	1	1	1	1	0
-0.53694	1.374495	0.93621	0.064694	0.856782	0.862423	0.65787	2.927123	-0.01755	-0.01441	0.177696	0.138446	-0.48085	-0.2081	0.08227	-0.21237	0.465435	-0.41917	0	1	1	1	1	0
-0.18436	-0.77999	-0.66197	-0.17159	0.031629	0.035668	-0.30478	-0.61203	-0.43112	-0.32481	-0.86648	0.473242	-0.75483	-0.83904	-0.12583	-0.2975	-0.44681	-0.35968	0	1	1	1	0	0
1.225943	-0.77999	0.93621	1.414869	0.36169	14.75191	-0.92363	-0.71315	1.51264	-2.08372	-1.50459	-1.56343	0.047538	1.432353	0.914684	0.936889	0.959571	0.096455	0	1	1	1	0	0
0.520791	-1.31861	-0.66197	-0.75385	0.36169	-0.12968	0.520349	0.399153	0.230586	-0.32481	-0.34439	-0.67064	0.800985	-0.71285	-0.12583	-0.25494	-0.21875	-0.12169	1	1	1	1	0	0
-1.24209	1.374495	0.93621	0.571009	0.36169	0.36637	-0.30478	-0.4098	-0.76197	0.114924	-0.11235	0.445342	-0.53956	0.927598	0.08227	-0.55289	-0.37079	-0.35968	0	1	1	1	1	0
-1.24209	-0.24137	-1.06152	-0.44162	-0.1334	0.035668	0.451588	0.095797	-0.88604	0.244256	-0.22837	-0.14055	0.849911	0.170466	-0.54204	-0.21237	-0.06671	0.413763	1	1	1	1	0	0
-0.53694	0.835874	0.13712	-0.10408	0.031629	0.035668	-0.64859	-0.4098	0.147873	-2.13546	-0.75046	-1.42393	-0.95053	1.116881	0.08227	-0.46776	-0.59886	-0.4985	0	2	1	1	0	0
-1.59467	-0.24137	0.13712	-0.69478	0.36169	-0.12968	0.245305	0.196916	-0.18298	-1.23013	-0.17036	-0.86594	-0.70591	-1.21761	-0.75014	-0.59546	-0.71289	-0.55799	1	1	1	1	1	0
-1.59467	0.835874	0.13712	-0.27285	0.36169	0.36637	-0.23602	-0.91539	-0.01755	-1.256	-0.34439	-0.78224	-0.76462	0.359749	0.08227	0.298411	-0.33278	-0.47867	0	1	1	1	1	0
0.168215	0.835874	1.7353	1.499255	-0.46346	-0.29503	-0.0985	-0.20756	-0.05891	-0.4024	-0.80847	-0.28005	0.360659	0.107372	0.706581	0.638933	0.959571	0.453427	0	1	1	1	1	0
0.873367	0.297252	-0.26243	0.317852	-1.12358	-0.46039	-0.16726	0.904747	0.189229	0.037325	-0.86648	-0.05685	0.879266	0.612126	0.290374	-0.2975	-0.21875	0.215446	0	1	1	1	1	1
-0.88951	0.835874	-1.06152	-1.2011	0.031629	-0.29503	-0.78611	-1.11763	-0.51383	-0.5576	0.525754	-0.41955	-0.8429	-1.3438	0.08227	-0.2975	-0.48482	-0.37951	1	1	1	1	1	0
-1.59467	1.374495	0.13712	0.064694	0.856782	0.862423	0.520349	-0.00532	-0.22434	0.295989	0.293715	0.58484	-0.73526	1.558542	0.706581	0.085585	0.161352	-0.10186	0	1	1	1	1	1

Logistic Regression-OvR

L1 Classification report:

	precision	recall	f1-score	support
0	0.57	0.23	0.33	42769
1	0.72	0.72	0.72	98675
micro avg	0.70	0.57	0.63	141444
macro avg	0.64	0.48	0.52	141444
weighted avg	0.67	0.57	0.60	141444
samples avg	0.35	0.31	0.32	141444

AUC Score: 0.6565195850159311

L1 Accuracy: 57.21%

L2 Accuracy: 57.20%

AUC Score: 0.6564345797253299

L2 Classification report:

	precision	recall	f1-score	support
0	0.57	0.23	0.33	42769
1	0.72	0.72	0.72	98675
micro avg	0.70	0.57	0.63	141444
macro avg	0.64	0.48	0.52	141444
weighted avg	0.67	0.57	0.60	141444
samples avg	0.35	0.31	0.32	141444

Independent response variables

```
Smoker model accuracy: 79.59%
Drinker model accuracy: 72.16%
AUC Score for drinking prediction:  0.7215461265197778
AUC Score for smoking prediction:  0.5915086861626121
Smoker model classification report:
              precision    recall  f1-score   support

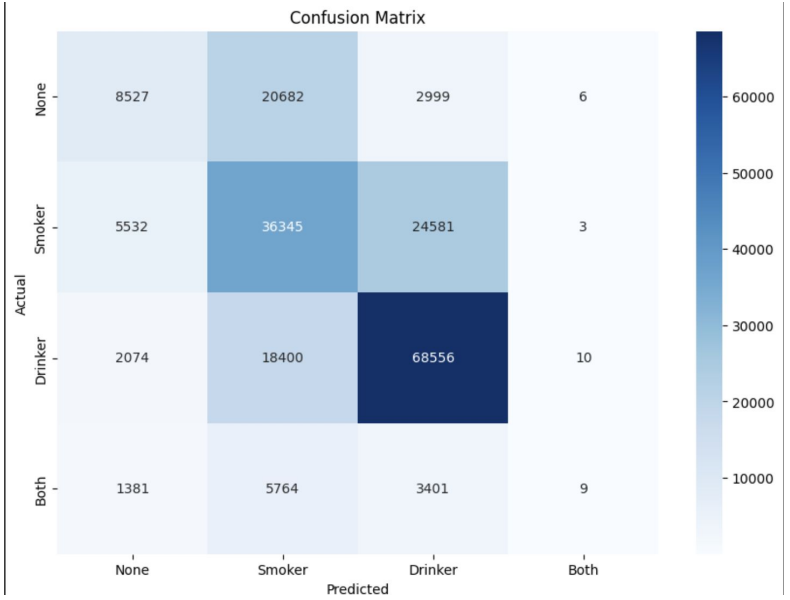
         0.0       0.82      0.95      0.88    155501
         1.0       0.57      0.23      0.33     42769

    accuracy                           0.80    198270
   macro avg       0.69      0.59      0.60    198270
weighted avg       0.76      0.80      0.76    198270

Drinker model classification report:
              precision    recall  f1-score   support

           0       0.72      0.72      0.72     99595
           1       0.72      0.72      0.72     98675

    accuracy                           0.72    198270
   macro avg       0.72      0.72      0.72    198270
weighted avg       0.72      0.72      0.72    198270
```



Decision Trees

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import GaussianNB, CategoricalNB
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score, confusion_matrix, f1_score, precision_score, recall_score, mean_squared_error, roc_auc_score, roc_curve

df = pd.read_csv('scaled_smoking_drinking_dataset.csv')

y_drinking = df.loc[:, 'DRK_YN']
y_smoking = df.loc[:, 'SMK_stat_type_cd']
X = df.drop(['DRK_YN', 'SMK_stat_type_cd'], axis=1)

X_train_drinking, X_test_drinking, y_train_drinking, y_test_drinking = train_test_split(X, y_drinking, test_size=0.25, random_state=21)
X_train_smoking, X_test_smoking, y_train_smoking, y_test_smoking = train_test_split(X, y_smoking, test_size=0.25, random_state=21)
```

```
# Decision Trees
max_depths = [2,4,6,8,10,12,14,16,18,20]

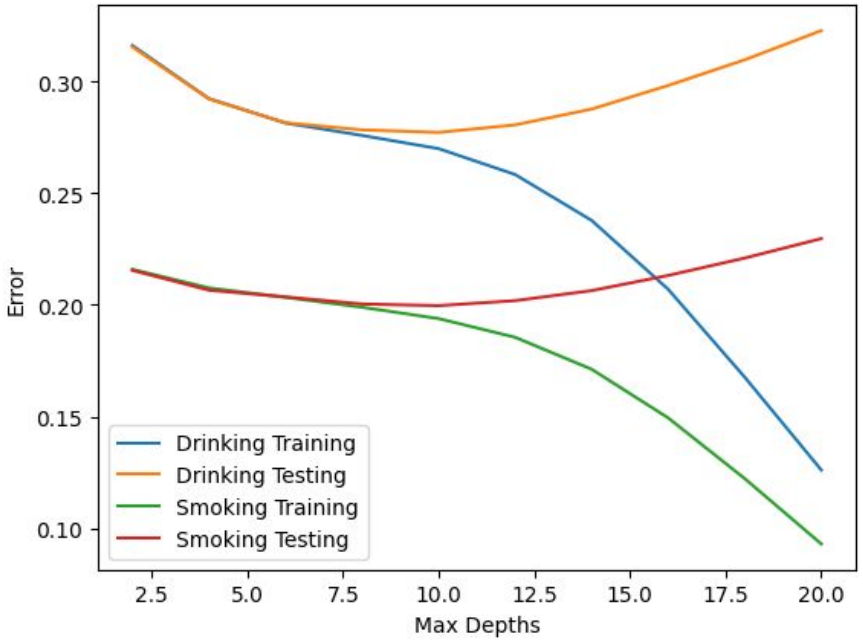
training_drinking_results = []
testing_drinking_results = []

training_smoking_results = []
testing_smoking_results = []

for max_depth in max_depths:
  dtreeD1 = DecisionTreeClassifier(max_depth=max_depth)
  dtreeD1 = dtreeD1.fit(X_train_drinking, y_train_drinking)
  y_predD1 = dtreeD1.predict(X_train_drinking)
  dtreeD2 = DecisionTreeClassifier(max_depth=max_depth)
  dtreeD2 = dtreeD2.fit(X_train_drinking, y_train_drinking)
  y_predD2 = dtreeD2.predict(X_test_drinking)
  training_drinking_results.append(1 - accuracy_score(y_train_drinking, y_predD1))
  testing_drinking_results.append(1 - accuracy_score(y_test_drinking, y_predD2))

  dtreeS1 = DecisionTreeClassifier(max_depth=max_depth)
  dtreeS1 = dtreeS1.fit(X_train_smoking, y_train_smoking)
  y_predS1 = dtreeS1.predict(X_train_smoking)
  dtreeS2 = DecisionTreeClassifier(max_depth=max_depth)
  dtreeS2 = dtreeS2.fit(X_train_smoking, y_train_smoking)
  y_predS2 = dtreeS2.predict(X_test_smoking)
  training_smoking_results.append(1 - accuracy_score(y_train_smoking, y_predS1))
  testing_smoking_results.append(1 - accuracy_score(y_test_smoking, y_predS2))

plt.plot(max_depths, training_drinking_results, label="Drinking Training")
plt.plot(max_depths, testing_drinking_results, label="Drinking Testing")
plt.plot(max_depths, training_smoking_results, label="Smoking Training")
plt.plot(max_depths, testing_smoking_results, label="Smoking Testing")
plt.legend(loc='best')
plt.xlabel("Max Depths")
plt.ylabel("Error")
plt.show()
```


0.30
0.25
0.20
0.15
0.10
Error
Drinking Training
Drinking Testing
Smoking Training
Smoking Testing
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Max Depths

```
dtreeD = DecisionTreeClassifier(max_depth=10)
dtreeD = dtreeD.fit(X_train_drinking, y_train_drinking)
y_predD1 = dtreeD.predict(X_train_drinking)
dtreeD = DecisionTreeClassifier(max_depth=10)
dtreeD = dtreeD.fit(X_train_drinking, y_train_drinking)
y_predD2 = dtreeD.predict(X_test_drinking)

dtreeS = DecisionTreeClassifier(max_depth=10)
dtreeS = dtreeS.fit(X_train_smoking, y_train_smoking)
y_predS1 = dtreeS.predict(X_train_smoking)
dtreeS = DecisionTreeClassifier(max_depth=10)
dtreeS = dtreeS.fit(X_train_smoking, y_train_smoking)
y_predS2 = dtreeS.predict(X_test_smoking)

print("Drinking:")
print("Training Set:")
print("Error: {}, Accuracy: {}, F1: {}, AUC: {}".format(1 - accuracy_score(y_train_drinking, y_predD1), accuracy_score(y_train_drinking, y_predD1),
                                                        f1_score(y_train_drinking, y_predD1), roc_auc_score(y_train_drinking, y_predD1)))
print("Test Set:")
print("Error: {}, Accuracy: {}, F1: {}, AUC: {}".format(1 - accuracy_score(y_test_drinking, y_predD2), accuracy_score(y_test_drinking, y_predD2),
                                                        f1_score(y_test_drinking, y_predD2), roc_auc_score(y_test_drinking, y_predD2)))


print("Smoking:")
print("Training Set:")
print("Error: {}, Accuracy: {}, F1: {}, AUC: {}".format(1 - accuracy_score(y_train_smoking, y_predS1), accuracy_score(y_train_smoking, y_predS1),
                                                        f1_score(y_train_smoking, y_predS1), roc_auc_score(y_train_smoking, y_predS1)))
print("Test Set:")
print("Error: {}, Accuracy: {}, F1: {}, AUC: {}".format(1 - accuracy_score(y_test_smoking, y_predS2), accuracy_score(y_test_smoking, y_predS2),
                                                        f1_score(y_test_smoking, y_predS2), roc_auc_score(y_test_smoking, y_predS2)))
```

```
Drinking:
Training Set:
Error: 0.2699832819777568, Accuracy: 0.7300167180222432, F1: 0.7336336675048599, AUC: 0.7300178369912973
Test Set:
Error: 0.27722656423375036, Accuracy: 0.7227734357662496, F1: 0.726827425799654, AUC: 0.7227930621521262
Smoking:
Training Set:
Error: 0.193873914101914, Accuracy: 0.806126085898086, F1: 0.4000249734656927, AUC: 0.6225077062531852
Test Set:
Error: 0.19964734886235713, Accuracy: 0.8003526511376429, F1: 0.38021394393366237, AUC: 0.6131576329399256
```

Naïve Bayes

```
gnb = GaussianNB()
gnb = gnb.fit(X_train_drinking, y_train_drinking)
y_pred = gnb.predict(X_test_drinking)
print("Naive Bayes - Drinking")
print("Testing Set: Accuracy: {}, Precision: {}, Recall: {}, F1: {}".format(accuracy_score(y_test_drinking, y_pred), precision_score(y_test_drinking, y_pred),
                                                                            recall_score(y_test_drinking, y_pred), f1_score(y_test_drinking, y_pred)))

gnb = GaussianNB()
gnb = gnb.fit(X_test_smoking, y_test_smoking)
y_pred = gnb.predict(X_test_smoking)
print("Naive Bayes - Smoking")
print("Testing Set: Accuracy: {}, Precision: {}, Recall: {}, F1: {}".format(accuracy_score(y_test_smoking, y_pred), precision_score(y_test_smoking, y_pred),
                                                                            recall_score(y_test_smoking, y_pred), f1_score(y_test_smoking, y_pred)))
```

```
Naive Bayes - Drinking
Testing Set: Accuracy: 0.6856603332028712, Precision: 0.6908485579924258, Recall: 0.6706473611066223, F1: 0.6805980919113276
Naive Bayes - Smoking
Testing Set: Accuracy: 0.7222488974608311, Precision: 0.4131088462188186, Recall: 0.686851664606973, F1: 0.5159176095808046
```

Neural Network

```
def build_ffnn_model():
    ffnn_model = tf.keras.Sequential([
      tf.keras.Input(shape=(X.shape[1],)),
      tf.keras.layers.Dense(22, activation=tf.nn.relu),
      tf.keras.layers.Dense(12, activation=tf.nn.relu),
      tf.keras.layers.Dense(2, activation=tf.nn.sigmoid)
    ])

    return ffnn_model
```

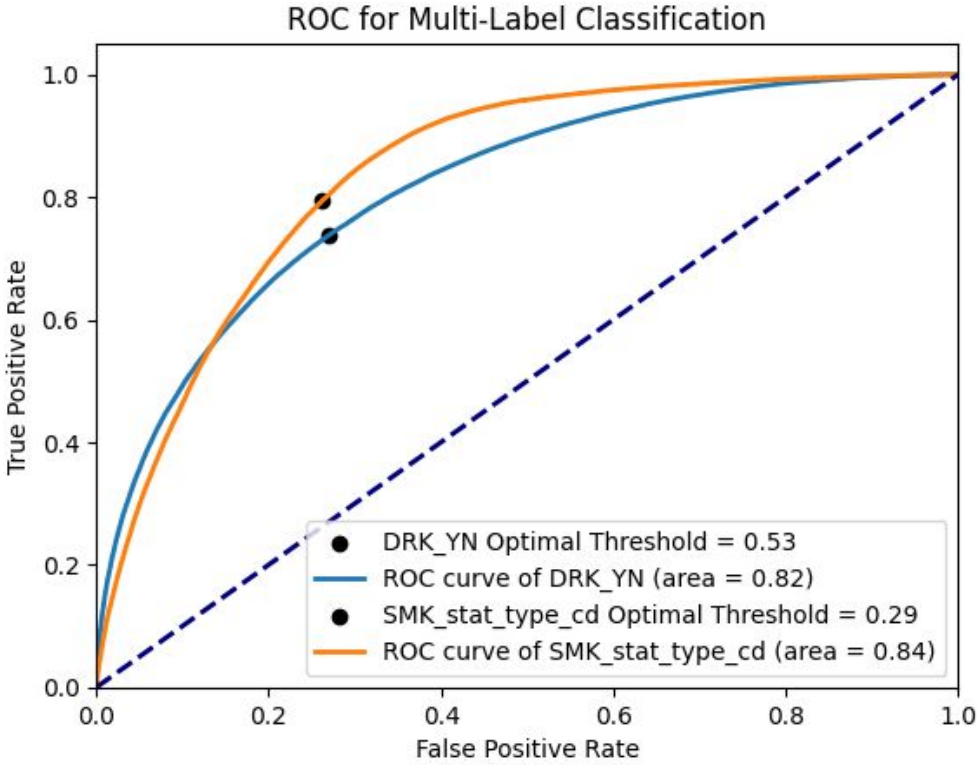
✓ 0.0s

```
model = build_ffnn_model()
```

✓ 0.0s

```
sgd = tf.keras.optimizers.SGD(learning_rate=0.01, momentum=0.9)
model.compile(loss='binary_crossentropy', optimizer=sgd, metrics=['accuracy'])
```

✓ 0.0s


ROC for Multi-Label Classification
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
DRK_YN Optimal Threshold = 0.53
ROC curve of DRK_YN (area = 0.82)
SMK_stat_type_cd Optimal Threshold = 0.29
ROC curve of SMK_stat_type_cd (area = 0.84)

	accuracy	precision	recall	f1	AUC
Drinker?	0.734333	0.729374	0.744998	0.737103	0.814997
Smoker?	0.746442	0.451435	0.807108	0.579014	0.840835

Which performed best?